1. Installation and setup of the java development kit (JDK), setup of Android SDK, setup of Eclipse IDE, setup of Android development tools (ADT) plugins, create an Android virtual device.

➢ Steps :-

1. **Install Java Development Kit (JDK):**
   - Download the latest version of JDK from the official Oracle website: https://www.oracle.com/java/technologies/javase-downloads.html
   - Run the installer and follow the instructions to complete the installation.
   - After the installation, set the PATH environment variable for Java on your computer

2. **Install Android SDK:**
   - Download the latest version of Android SDK Command-line Tools from the official Android website: https://developer.android.com/studio#downloads
   - Extract the downloaded file to a directory on your computer.
   - Set the ANDROID_HOME environment variable to the directory where you extracted the Android SDK. To do this, follow the instructions for your operating system:
   - Windows: https://developer.android.com/studio/intro/update#set-upyour- android-home-environment-variable
   - Mac: https://developer.android.com/studio/intro/update#macos-usr-local

3. **Install Eclipse IDE:**
   - Download the latest version of Eclipse IDE for Java Developers from the official Eclipse website: https://www.eclipse.org/downloads/packages/
   - Extract the downloaded file to a directory on your computer.

4. **Install Android Development Tools (ADT) plugins:**
   - Open Eclipse IDE.
   - Click on Help > Eclipse Marketplace.
   - Search for "Android Development Tools" and click "Go".
   - Select "Android Development Tools for Eclipse" and click "Install".

- Follow the instructions to complete the installation.
- Restart Eclipse IDE.

5. **Create an Android virtual device:**
    - Open Eclipse IDE.
    - Click on Window > Android SDK and AVD Manager.
    - Click on the "Virtual Devices" tab.
    - Click the "New" button to create a new virtual device.
    - Follow the instructions to configure the virtual device.
    - Click the "Create AVD" button to create the virtual device.

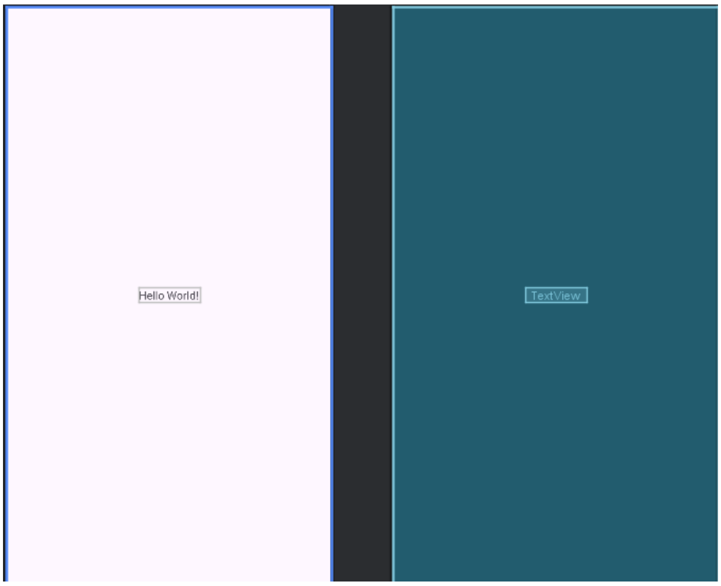
That's it! We have now installed and set up Java Development Kit (JDK), Android SDK, Eclipse IDE, Android Development Tools (ADT) plugins, and created an Android virtual device. We can now start developing Android applications in Eclipse IDE.

2. Develop a Program to display Hello World on Screen.

- **Program :-**

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
       android:textSize="50dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```



MainActivity.java

```
package com.example.myapplication;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

- **Output :-**

3. Develop a program to implement Linear Layout, Absolute Layout and Relative Layout.

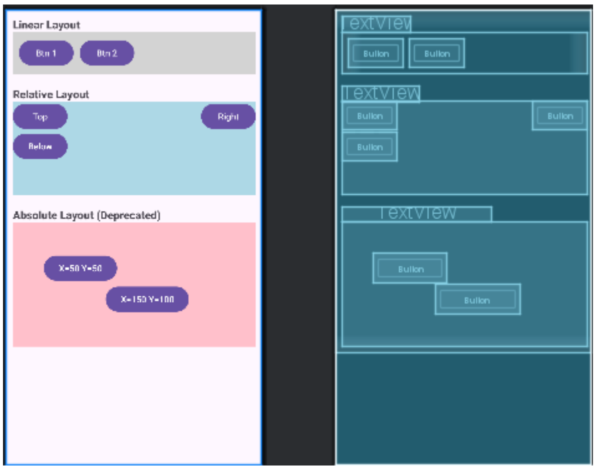
- **Program :-**

Activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="10dp">

        <!-- 🟢 Linear Layout Section -->
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Linear Layout"
            android:textStyle="bold"
            android:textSize="18sp"/>

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal"
            android:background="#D3D3D3"
            android:padding="10dp">

            <Button
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Btn 1"/>

            <Button
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Btn 2"
                android:layout_marginStart="10dp"/>

        </LinearLayout>
```

```
<!-- □ Relative Layout Section -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Relative Layout"
    android:textStyle="bold"
    android:textSize="18sp"
    android:layout_marginTop="20dp"/>

<RelativeLayout
    android:layout_width="match_parent"
    android:layout_height="150dp"
    android:background="#ADD8E6">

    <Button
        android:id="@+id/topButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Top"/>

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Below"
        android:layout_below="@id/topButton"/>

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Right"
        android:layout_alignParentEnd="true"/>

</RelativeLayout>


<!-- ● Absolute Layout Section (Deprecated) -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Absolute Layout (Deprecated)"
    android:textStyle="bold"
    android:textSize="18sp"
    android:layout_marginTop="20dp"/>

<AbsoluteLayout
    android:layout_width="match_parent"
```

```
        android:layout_height="200dp"
        android:background="#FFC0CB">

        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="X=50 Y=50"
            android:layout_x="50dp"
            android:layout_y="50dp"/>

        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="X=150 Y=100"
            android:layout_x="150dp"
            android:layout_y="100dp"/>

    </AbsoluteLayout>

  </LinearLayout>
</ScrollView>
```

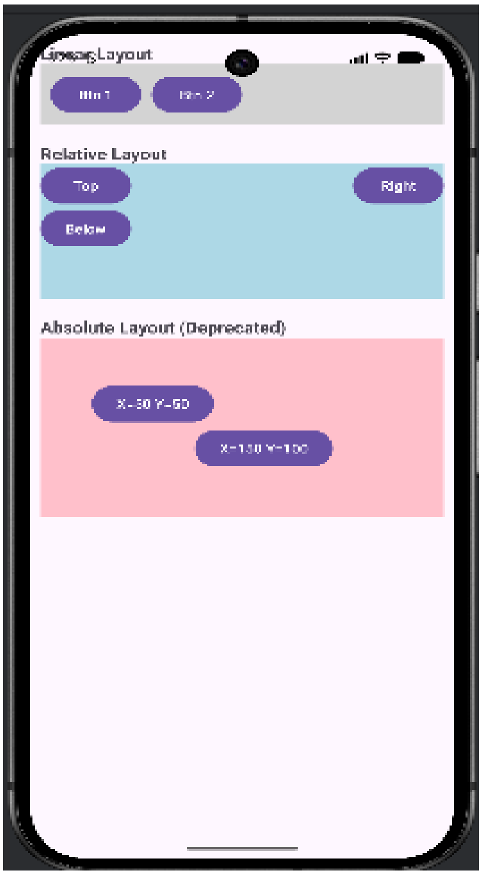


MainActivity.java

```
package com.example.layouts_;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
```

```
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);

    }
}
```

- **Output :-**

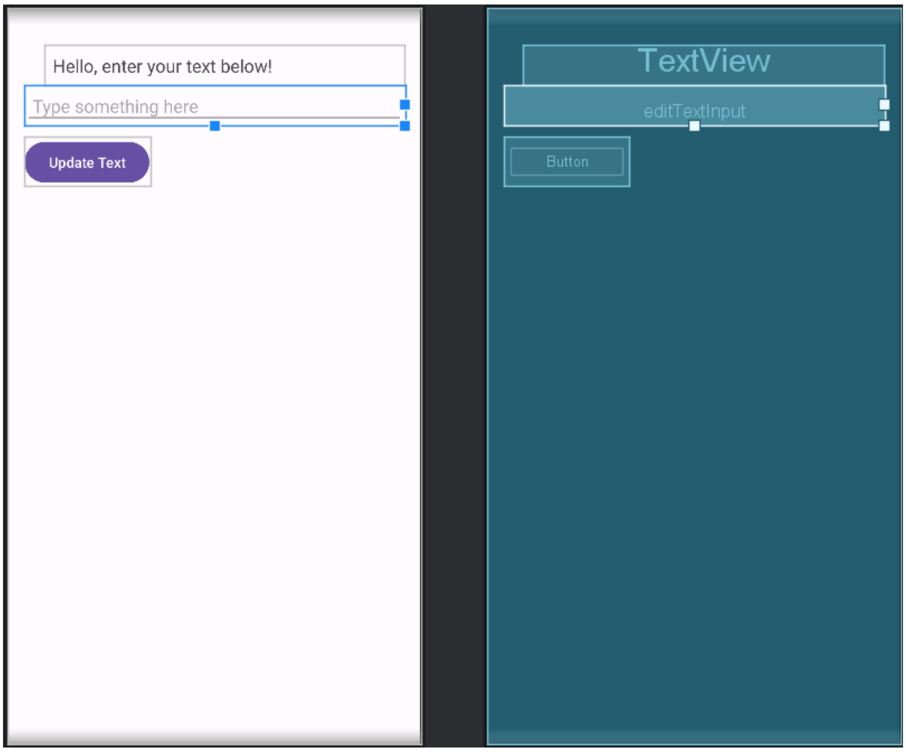


4. Develop a program to implement TextView and EditText.

- **Program :-**

Layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <!-- Display text -->
    <TextView
        android:id="@+id/textViewDisplay"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginStart="20dp"
        android:layout_marginTop="20dp"
        android:padding="8dp"
        android:text="Hello, enter your text below!"
        android:textSize="18sp" />

    <!-- Input field -->
    <EditText
        android:id="@+id/editTextInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Type something here"
        android:inputType="textPersonName"
        android:padding="8dp" />

    <!-- Button to update TextView -->
    <Button
        android:id="@+id/buttonUpdate"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Update Text"
        android:layout_marginTop="12dp" />

</LinearLayout>
```



Java/com.example.appname/MainActivity.java

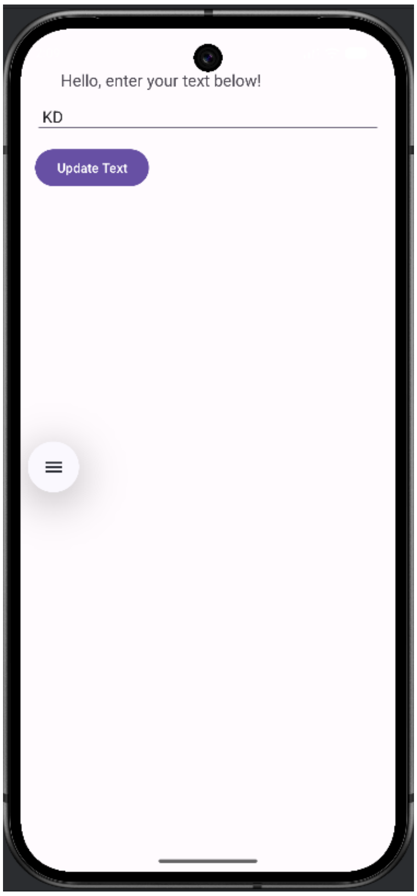
```
package com.example.p4;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;

import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    private TextView textViewDisplay;
    private EditText editTextInput;
    private Button buttonUpdate;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize UI components
        textViewDisplay = findViewById(R.id.textViewDisplay);
        editTextInput = findViewById(R.id.editTextInput);
        buttonUpdate = findViewById(R.id.buttonUpdate);
        // Button click listener
```

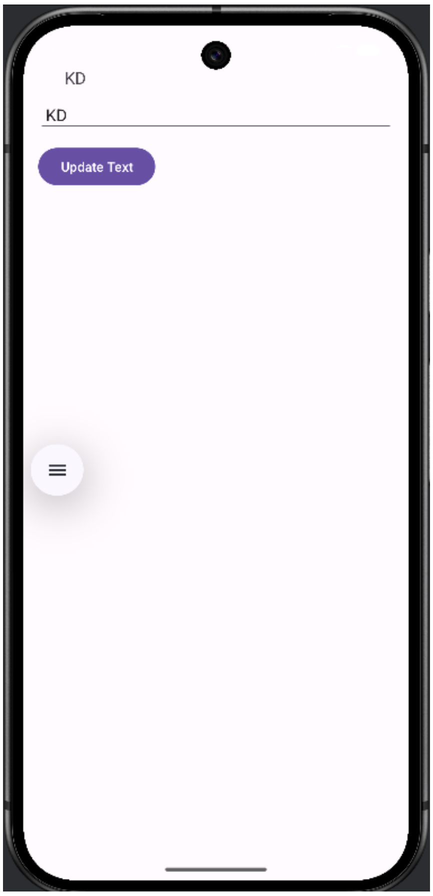
```
        buttonUpdate.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String userInput = editTextInput.getText().toString();
                if (!userInput.isEmpty()) {
                    textViewDisplay.setText(userInput);
                } else {
                    textViewDisplay.setText("Please enter some text!");
                }
            }
        });

    }
}
```

Res/values/string.xml

```
<resources>
    <string name="app_name">p4</string>
    <string name="text_to_show">Hello, enter your text below_2</string>
</resources>
```

- **Output :-**





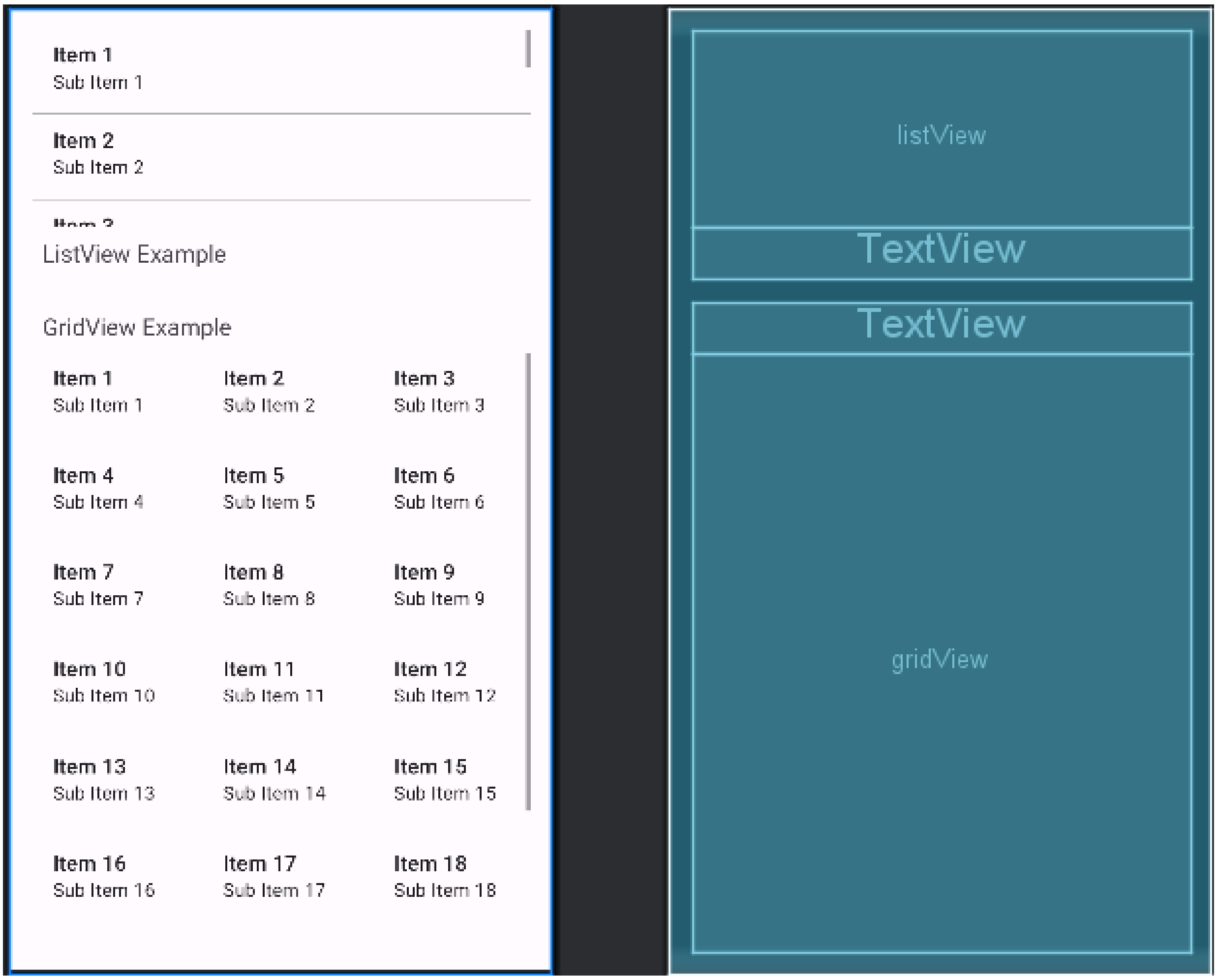
5. Develop a program to implement List View, Grid View.

- **Program :-**

Layout/activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <!-- ListView Section -->

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="150dp"
        android:divider="@android:color/darker_gray"
        android:dividerHeight="1dp" />

    <TextView
        android:id="@+id/textViewList"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="ListView Example"
        android:textSize="18sp"
        android:padding="8dp" />

    <!-- GridView Section -->

    <TextView
        android:id="@+id/textViewGrid"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="GridView Example"
        android:textSize="18sp"
        android:padding="8dp"
        android:layout_marginTop="16dp" />

    <GridView
        android:id="@+id/gridView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:numColumns="3"
        android:horizontalSpacing="10dp"
        android:verticalSpacing="10dp"
```

```
        android:stretchMode="columnWidth"
        android:gravity="center" />

</LinearLayout>
```



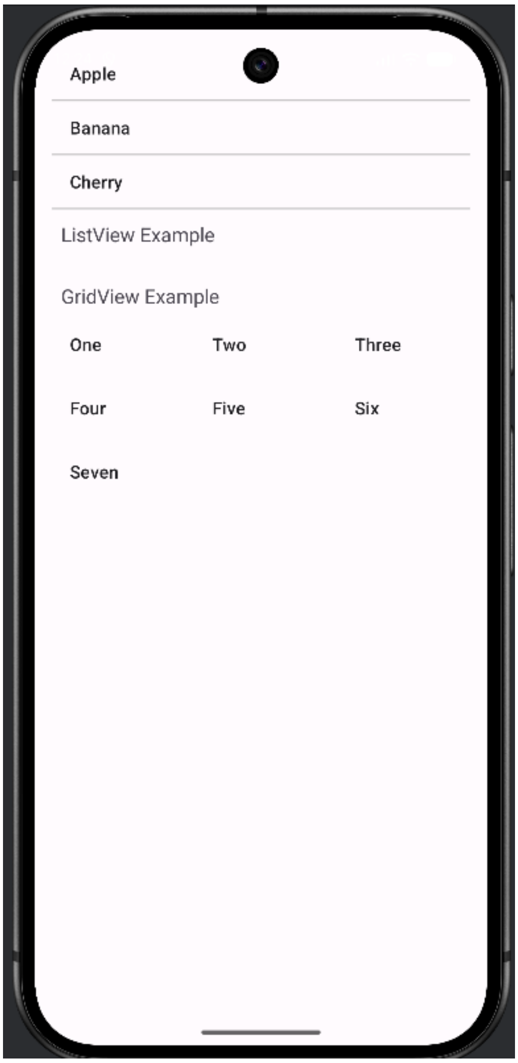
Java/com.example.appname/MainActivity.java

```
package com.example.p5;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.GridView;
import android.widget.ListView;
```

```
import android.widget.Toast;
import android.view.View;

public class MainActivity extends AppCompatActivity {

    private ListView listView;
    private GridView gridView;

    // Sample data
    String[] listItems = {"Apple", "Banana", "Cherry", "Date", "Elderberry"};
    String[] gridItems = {"One", "Two", "Three", "Four", "Five", "Six", "Seven"};

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        listView = findViewById(R.id.listView);
        gridView = findViewById(R.id.gridView);

        // Adapter for ListView
        ArrayAdapter<String> listAdapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                listItems
        );
        listView.setAdapter(listAdapter);

        // Adapter for GridView
        ArrayAdapter<String> gridAdapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                gridItems
        );
        gridView.setAdapter(gridAdapter);

        // Click listeners
        listView.setOnItemClickListener((parent, view, position, id) ->
                Toast.makeText(MainActivity.this,
                        "ListView clicked: " + listItems[position],
                        Toast.LENGTH_SHORT).show());

        gridView.setOnItemClickListener((parent, view, position, id) ->
                Toast.makeText(MainActivity.this,
                        "GridView clicked: " + gridItems[position],
                        Toast.LENGTH_SHORT).show());
```

```
    }
}
```

- **Output :-**


Apple
Banana
Cherry
ListView Example
GridView Example
One
Two
Three
Four
Five
Six
Seven

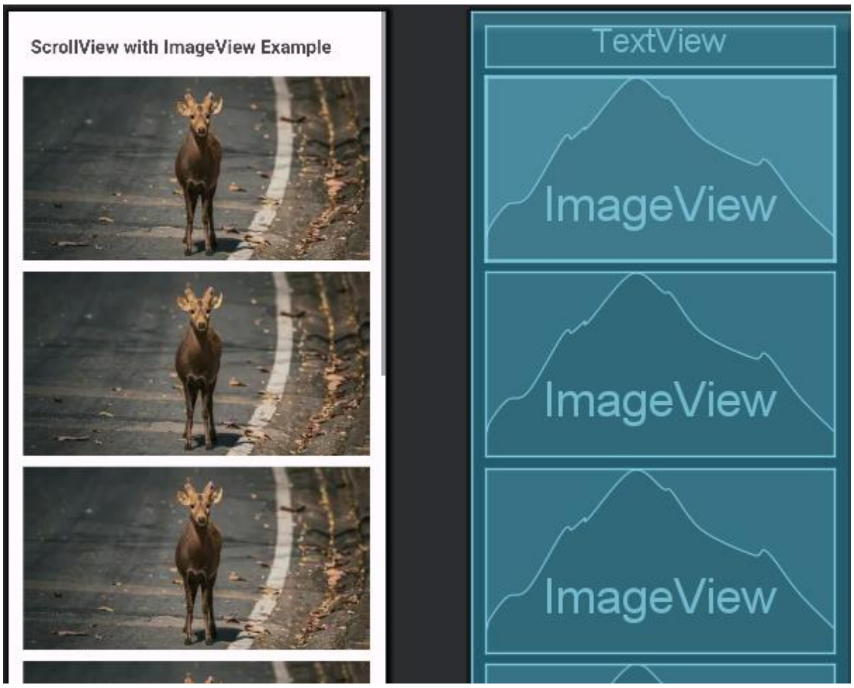
6. Develop a program to implement Image view and scroll view.

- **Program :-**

Layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:fillViewport="true">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <!-- Title -->
        <TextView
            android:id="@+id/textViewTitle"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="ScrollView with ImageView Example"
            android:textSize="20sp"
            android:textStyle="bold"
            android:padding="8dp" />

        <ImageView
            android:id="@+id/imageView1"
            android:layout_width="match_parent"
            android:layout_height="200dp"
            android:layout_marginTop="12dp"
            android:scaleType="centerCrop"
            android:src="@drawable/sample_image1" />

        <ImageView
            android:id="@+id/imageView2"
            android:layout_width="match_parent"
            android:layout_height="200dp"
            android:layout_marginTop="12dp"
            android:scaleType="centerCrop"
            android:src="@drawable/sample_image2" />

        <ImageView
            android:id="@+id/imageView3"
            android:layout_width="match_parent"
            android:layout_height="200dp"
```

```
        android:layout_marginTop="12dp"
        android:scaleType="centerCrop"
        android:src="@drawable/sample_image3" />

    <ImageView
        android:id="@+id/imageView3"
        android:layout_width="match_parent"
        android:layout_height="200dp"
        android:src="@drawable/sample_image3"
        android:scaleType="centerCrop"
        android:layout_marginTop="12dp" />

    <ImageView
        android:id="@+id/imageView3"
        android:layout_width="match_parent"
        android:layout_height="200dp"
        android:src="@drawable/sample_image3"
        android:scaleType="centerCrop"
        android:layout_marginTop="12dp" />

    <ImageView
        android:id="@+id/imageView3"
        android:layout_width="match_parent"
        android:layout_height="200dp"
        android:src="@drawable/sample_image3"
        android:scaleType="centerCrop"
        android:layout_marginTop="12dp" />
  </LinearLayout>
</ScrollView>
```



Java/com.example.appname/MainActivity.java

```java
package com.example.p6;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.ImageView;

public class MainActivity extends AppCompatActivity {

    private ImageView imageView1, imageView2, imageView3;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        imageView1 = findViewById(R.id.imageView1);
        imageView2 = findViewById(R.id.imageView2);
        imageView3 = findViewById(R.id.imageView3);

        // Optional: dynamically change images if needed
        imageView1.setImageResource(R.drawable.sample_image1);
        imageView2.setImageResource(R.drawable.sample_image2);
        imageView3.setImageResource(R.drawable.sample_image3);
    }
}
```

Res/drawable/string.xml

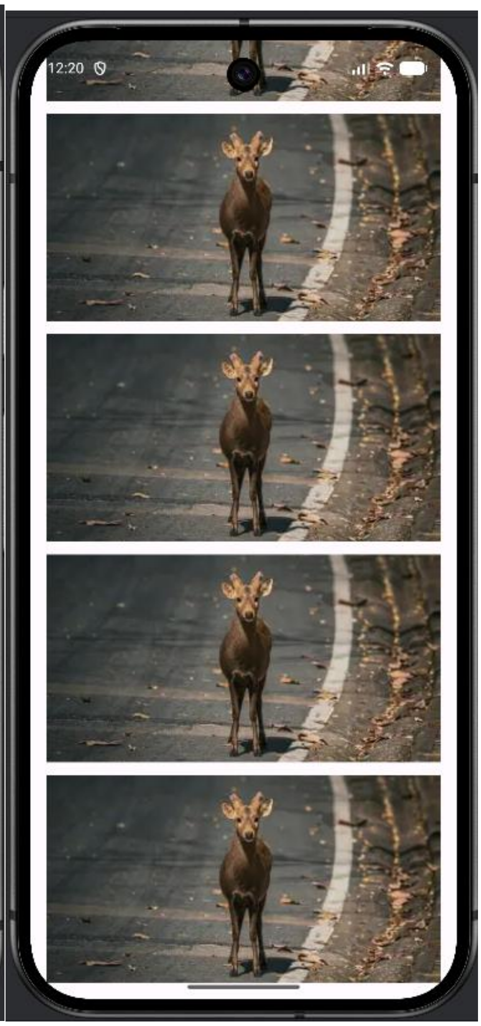
```xml
<resources>
    <string name="app_name">p6</string>
</resources>
```

Res/values/themes/themes.xml

```xml
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.P6" parent="Theme.Material3.DayNight.NoActionBar">
        <!-- Customize your light theme here. -->
        <!-- <item name="colorPrimary">@color/my_light_primary</item> -->
    </style>

    <style name="Theme.P6" parent="Base.Theme.P6" />
</resources>
```

Res/drawable:sample_image1.png, sample_image2.png, sample_image3.png

➢ Output :-

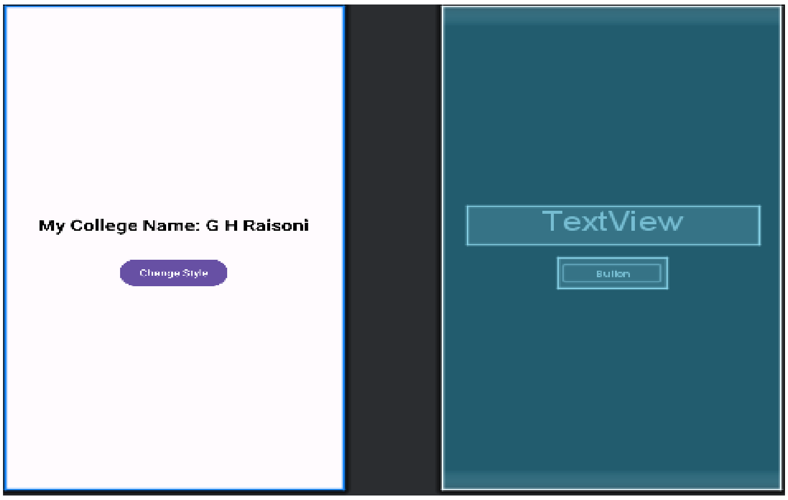


7. Create an Android Application that will change color of the College Name on click of Push Button and change the font size, font style of text view using xml.

- **Program :-**

Layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <!-- College Name TextView -->
    <TextView
        android:id="@+id/textViewCollege"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/college_name"
        android:textSize="24sp"
        android:textStyle="bold"
        android:textColor="@android:color/black"
        android:padding="12dp" />

    <!-- Push Button -->
    <Button
        android:id="@+id/buttonChange"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Change Style"
        android:layout_marginTop="20dp" />
</LinearLayout>
```



Java/com.example.appname/MainActivity.java

```
package com.example.p7;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import android.view.View;
import android.graphics.Color;
import android.graphics.Typeface;

public class MainActivity extends AppCompatActivity {

    private TextView textViewCollege;
    private Button buttonChange;
    private boolean isStyleOne = true;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        textViewCollege = findViewById(R.id.textViewCollege);
        buttonChange = findViewById(R.id.buttonChange);

        buttonChange.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if (isStyleOne) {
                    // Change to Style 2
                    textViewCollege.setTextColor(Color.RED);
                    textViewCollege.setTextSize(30); // bigger font
                    textViewCollege.setTypeface(Typeface.SERIF, Typeface.ITALIC);
                    isStyleOne = false;
                } else {
                    // Change back to Style 1
                    textViewCollege.setTextColor(Color.BLUE);
                    textViewCollege.setTextSize(24); // original size
                    textViewCollege.setTypeface(Typeface.SANS_SERIF, Typeface.BOLD);
                    isStyleOne = true;
                }
            }
        });
    }
}
```

Res/string.xml

```
<resources>
    <string name="app_name">p7</string>
    <string name="college_name">My College Name: G H Raisoni</string>
</resources>
```

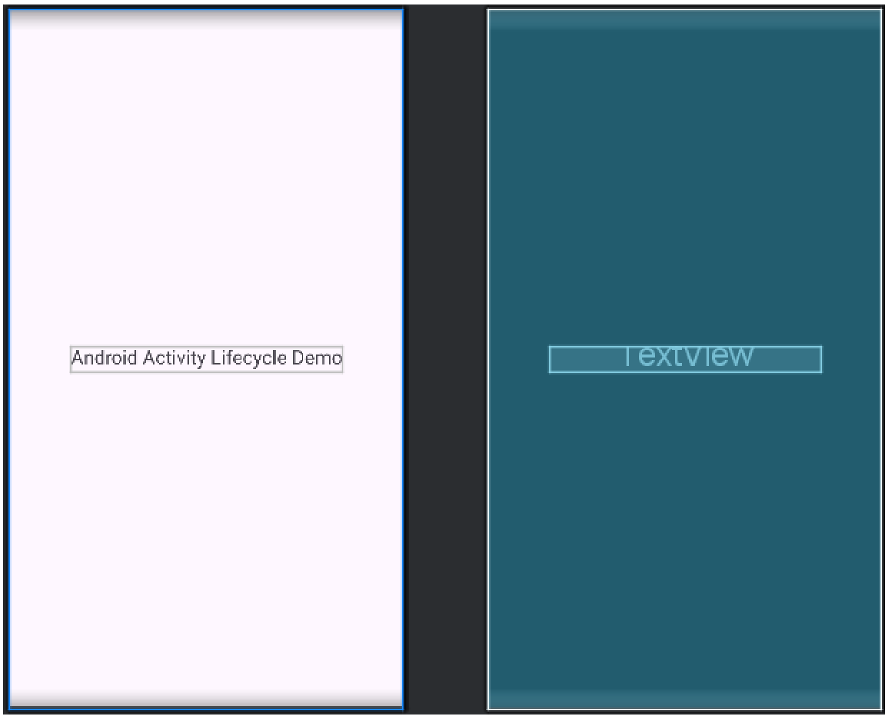
- **Output :-**

8. Create a Simple Application Which demonstrate Life Cycle of Activity.

- **Program :-**

Layout/activity_main.xml

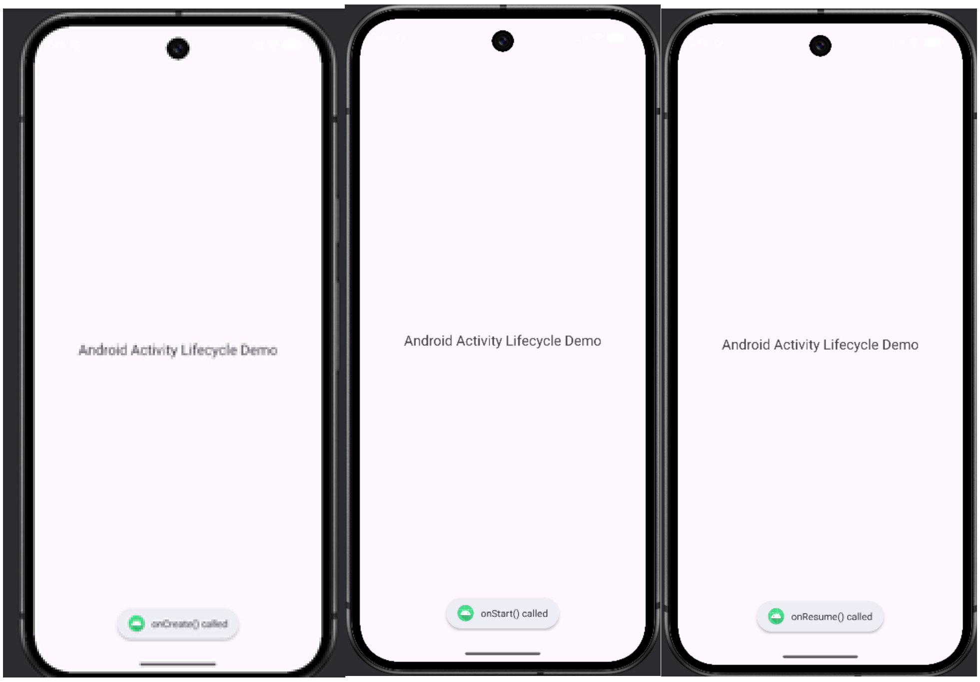
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"

    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Android Activity Lifecycle Demo"
        android:textSize="20sp"/>

</LinearLayout>
```



Java/com.example.appname/MainActivity.java

```java
package com.example.activitylifecycledemo;

import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;

import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    // Called when the activity is first created
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Toast.makeText(this, "onCreate() called", Toast.LENGTH_SHORT).show();
    }

    // Called when the activity becomes visible to the user
    @Override
    protected void onStart() {
        super.onStart();
        Toast.makeText(this, "onStart() called", Toast.LENGTH_SHORT).show();
    }

    // Called when the activity starts interacting with the user
    @Override
    protected void onResume() {
        super.onResume();
        Toast.makeText(this, "onResume() called", Toast.LENGTH_SHORT).show();
    }

    // Called when another activity partially covers this activity
    @Override
    protected void onPause() {
        super.onPause();
        Toast.makeText(this, "onPause() called", Toast.LENGTH_SHORT).show();
    }

    // Called when the activity is no longer visible
    @Override
    protected void onStop() {
        super.onStop();
        Toast.makeText(this, "onStop() called", Toast.LENGTH_SHORT).show();
    }

    // Called before the activity is destroyed
    @Override
    protected void onDestroy() {
        super.onDestroy();
        Toast.makeText(this, "onDestroy() called", Toast.LENGTH_SHORT).show();
    }

    // Called when the activity is restarting after being stopped
    @Override
    protected void onRestart() {
        super.onRestart();
        Toast.makeText(this, "onRestart() called", Toast.LENGTH_SHORT).show();
    }
}
```

➢ Output :-


Android Activity Lifecycle Demo
onCreate() called
Android Activity Lifecycle Demo
onStart() called
Android Activity Lifecycle Demo
onResume() called

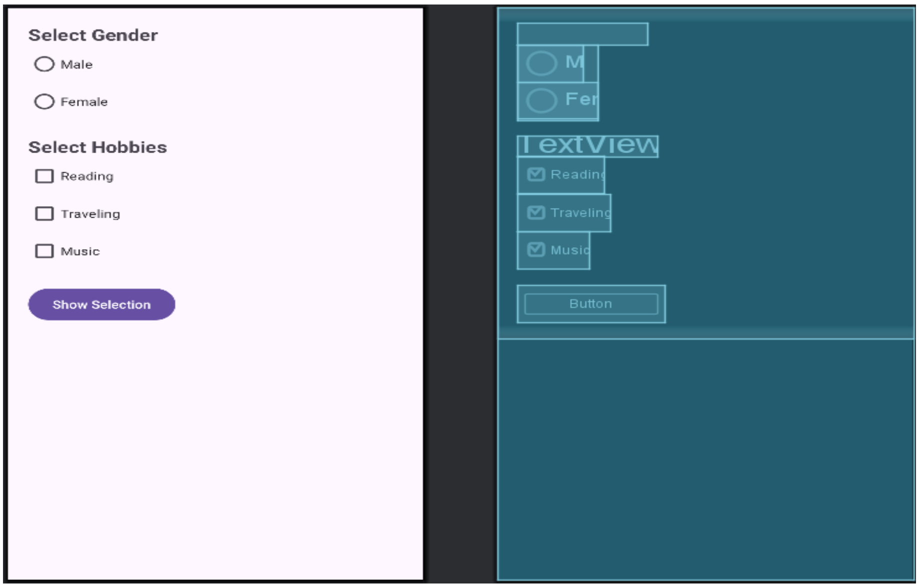
9. Design Following Screens Using Radio Buttons & Checkboxes. Display the selected text using Toast.

- **Program :-**

Layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="20dp">

        <!-- Radio Buttons Section -->
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Select Gender"
            android:textSize="20sp"
            android:textStyle="bold"/>

        <RadioGroup
            android:id="@+id/radioGroup"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content">

            <RadioButton
                android:id="@+id/radioMale"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Male"/>

            <RadioButton
                android:id="@+id/radioFemale"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Female"/>

        </RadioGroup>
        <!-- Checkbox Section -->
        <TextView
            android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="Select Hobbies"
        android:textSize="20sp"
        android:textStyle="bold"
        android:layout_marginTop="20dp"/>
    <CheckBox
        android:id="@+id/checkReading"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Reading"/>
    <CheckBox
        android:id="@+id/checkTravel"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Traveling"/>

    <CheckBox
        android:id="@+id/checkMusic"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Music"/>

    <!-- Button -->
    <Button
        android:id="@+id/btnSubmit"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show Selection"
        android:layout_marginTop="20dp"/>

  </LinearLayout>
</ScrollView>
```

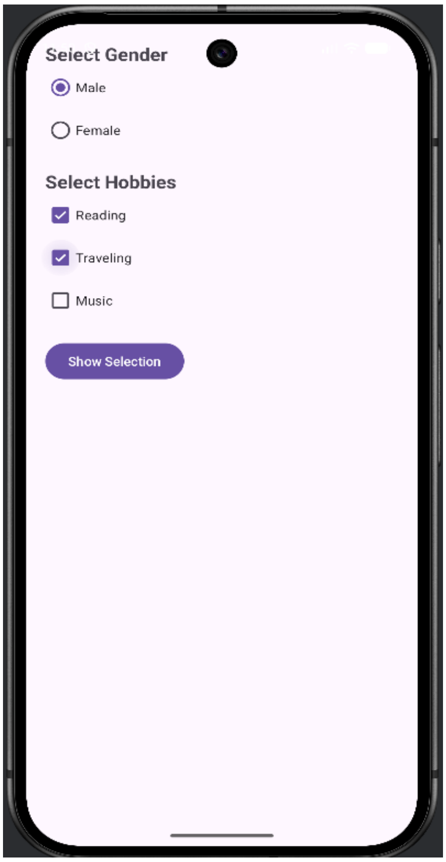


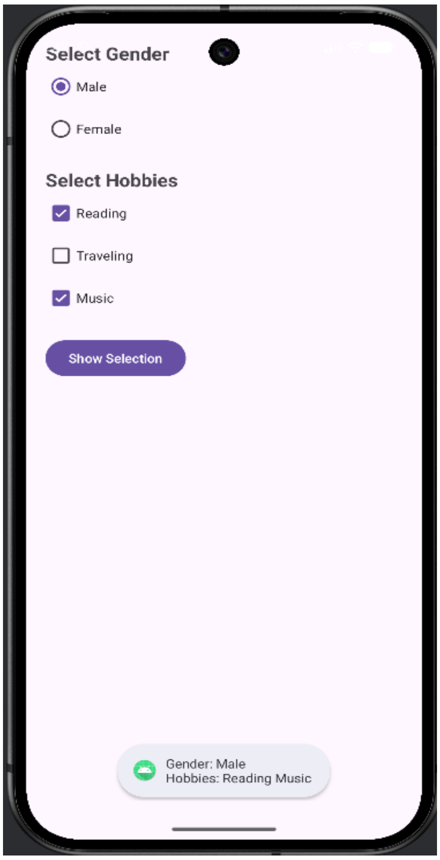
Java/com.example.appname/MainActivity.java

```
package com.example.practical_no_09;

import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.Toast;


public class MainActivity extends AppCompatActivity {

    RadioGroup radioGroup;
    RadioButton radioButton;

    CheckBox checkReading, checkTravel, checkMusic;

    Button btnSubmit;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        radioGroup = findViewById(R.id.radioGroup);

        checkReading = findViewById(R.id.checkReading);
        checkTravel = findViewById(R.id.checkTravel);
        checkMusic = findViewById(R.id.checkMusic);

        btnSubmit = findViewById(R.id.btnSubmit);

        btnSubmit.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {

                String result = "";

                // Radio Button selection
                int selectedId = radioGroup.getCheckedRadioButtonId();
                radioButton = findViewById(selectedId);
```

```
                if(radioButton != null){
                    result += "Gender: " + radioButton.getText().toString() + "\n";
                }
                // Checkbox selections
                result += "Hobbies: ";
                if(checkReading.isChecked()){
                    result += "Reading ";
                }
                if(checkTravel.isChecked()){
                    result += "Traveling ";
                }
                if(checkMusic.isChecked()){
                    result += "Music ";
                }
                // Show Toast
                Toast.makeText(MainActivity.this, result, Toast.LENGTH_LONG).show();
            }
        });
    }
}
```

➢ **Output :-**





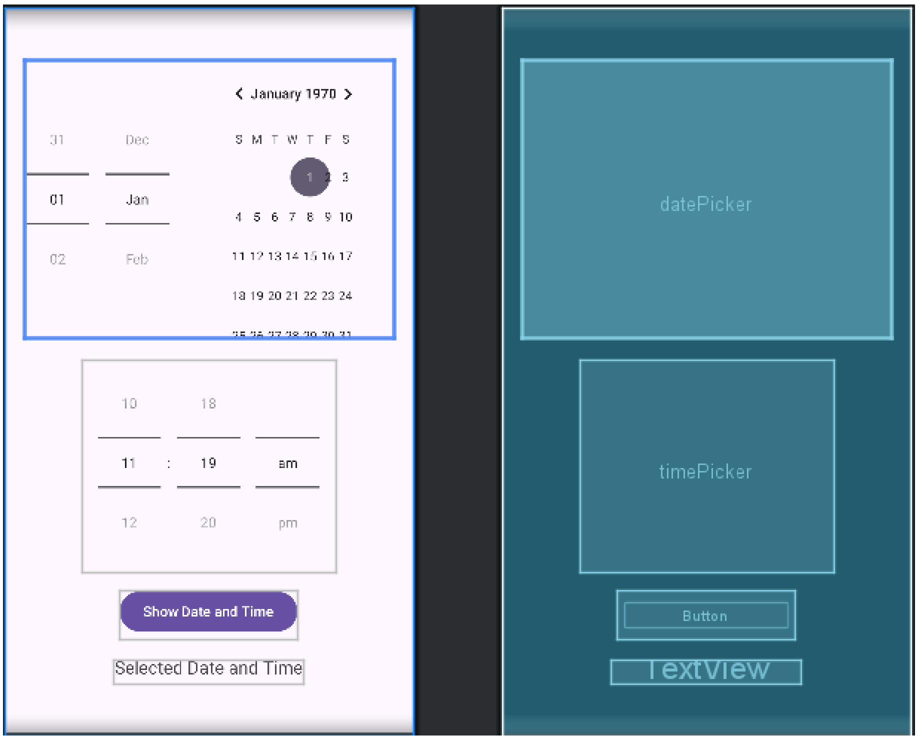
10. Develop a program to implement Date and time picker.

- **Program :-**

Layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center"

    tools:context=".MainActivity">

    <!-- Date Picker in Spinner Mode -->
    <DatePicker
        android:id="@+id/datePicker"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:datePickerMode="spinner"/>

    <!-- Time Picker in Spinner Mode -->
    <TimePicker
        android:id="@+id/timePicker"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:timePickerMode="spinner"
        android:layout_marginTop="20dp"/>

    <!-- Button -->
    <Button
        android:id="@+id/btnShow"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show Date and Time"
        android:layout_marginTop="20dp"/>

    <!-- Result Text -->
    <TextView
        android:id="@+id/txtResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Selected Date and Time"
```

```
        android:textSize="18sp"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

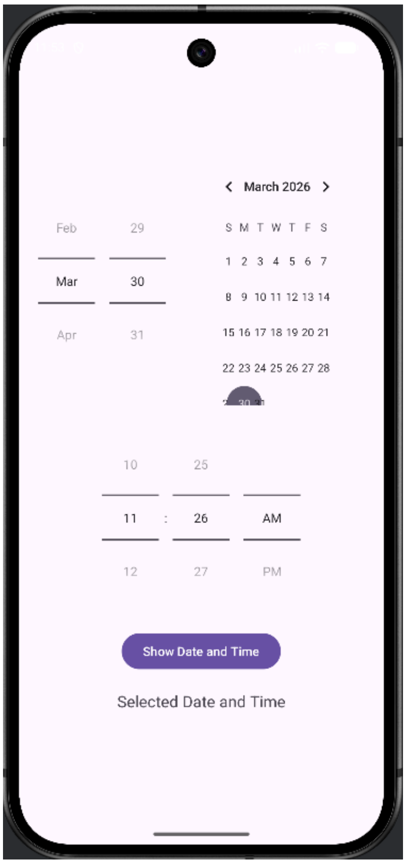


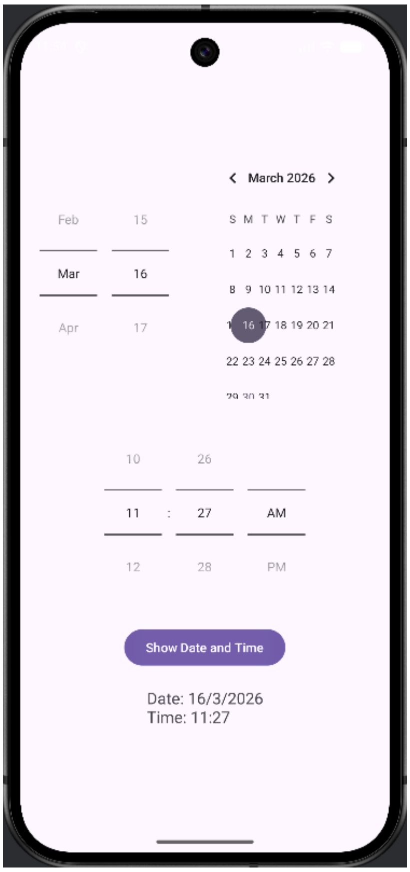
Java/com.example.appname/MainActivity.java

```
package com.example.datetimepicker3;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.DatePicker;
import android.widget.TextView;
import android.widget.TimePicker;

public class MainActivity extends AppCompatActivity {

    DatePicker datePicker;
    TimePicker timePicker;
    Button btnShow;
    TextView txtResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```

```
        datePicker = findViewById(R.id.datePicker);
        timePicker = findViewById(R.id.timePicker);
        btnShow = findViewById(R.id.btnShow);
        txtResult = findViewById(R.id.txtResult);

        btnShow.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int day = datePicker.getDayOfMonth();
                int month = datePicker.getMonth() + 1;
                int year = datePicker.getYear();
                int hour = timePicker.getCurrentHour();
                int minute = timePicker.getCurrentMinute();

                String result = "Date: " + day + "/" + month + "/" + year +
                        "\nTime: " + hour + ":" + minute;

                txtResult.setText(result);
            }
        });
    }
}
```

➢ **Output :-**



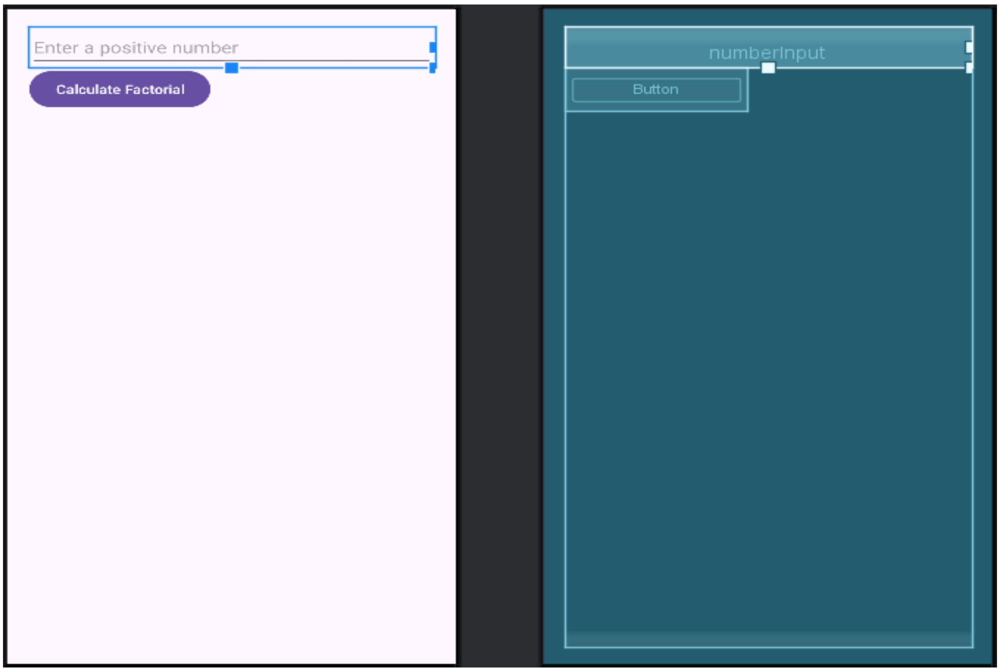


11. Create a Simple Application, which read a positive number from the user and Display its factorial value in another activity.

- **Program :-**

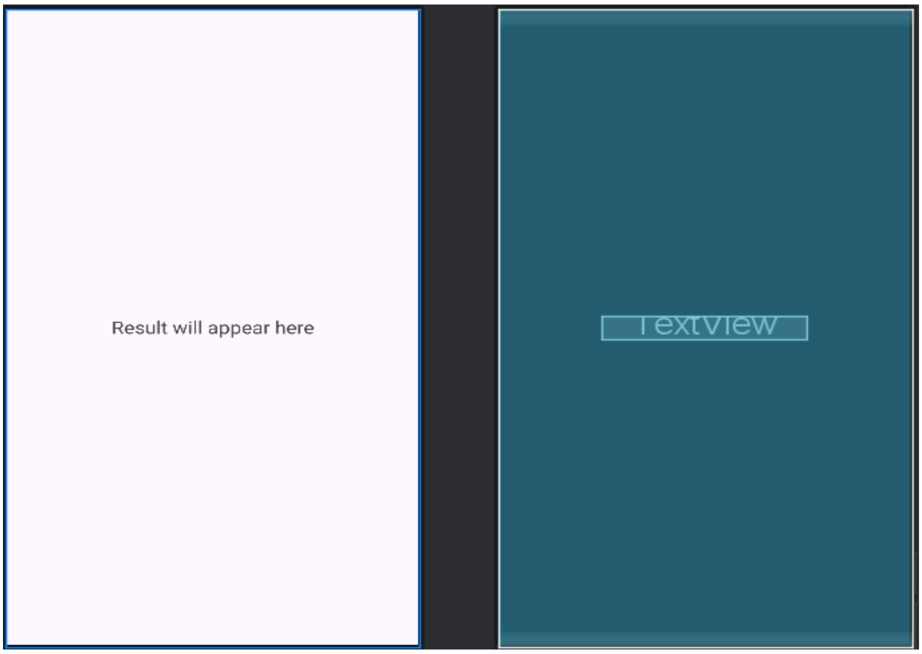
Layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="20dp"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/numberInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter a positive number"
        android:inputType="number" />
    <Button
        android:id="@+id/calcButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Calculate Factorial" />

</LinearLayout>
```



Layout/activity_result.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <TextView
        android:id="@+id/resultText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp"
        android:text="Result will appear here" />
</LinearLayout>
```



Manifest/AndroidMainfest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
```

```
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Factorialapp">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        <activity android:name=".ResultActivity"/>

    </application>

</manifest>
```

Java/com.example.appname/MainActivity.java

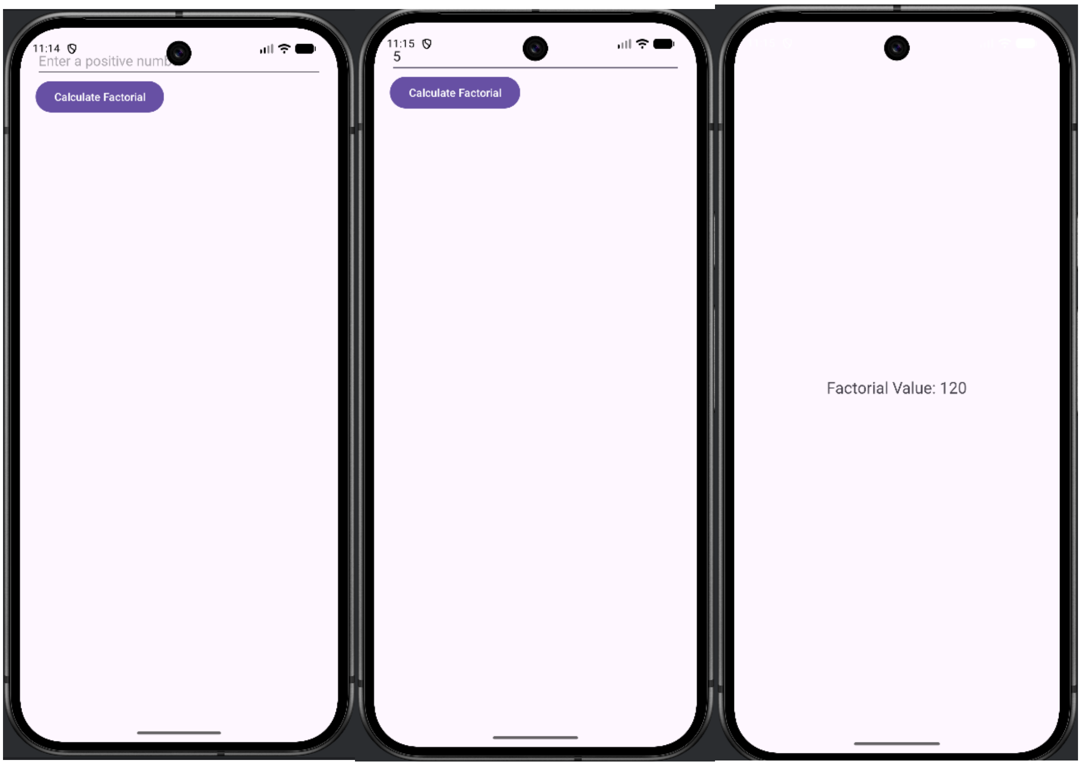
```
package com.example.factorialapp;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText numberInput;
    Button calcButton;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        numberInput = findViewById(R.id.numberInput);
        calcButton = findViewById(R.id.calcButton);

        calcButton.setOnClickListener(v -> {
            String input = numberInput.getText().toString();
            if (!input.isEmpty()) {
```

```java
            int num = Integer.parseInt(input);
            long factorial = calculateFactorial(num);

            // Pass result to second activity
            Intent intent = new Intent(MainActivity.this, ResultActivity.class);
            intent.putExtra("factorialResult", factorial);
            startActivity(intent);
        }
    });
}

private long calculateFactorial(int n) {
    long result = 1;
    for (int i = 1; i <= n; i++) {
        result *= i;
    }
    return result;
}
}
```

Java/com.example.appname/ResultActivity.java

```java
import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class ResultActivity extends AppCompatActivity {

    TextView resultText;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_result);

        resultText = findViewById(R.id.resultText);

        // Get factorial result from intent
        long factorial = getIntent().getLongExtra("factorialResult", 1);
        resultText.setText("Factorial Value: " + factorial);
    }
}
```

➢ Output :-


11:14
Enter a positive numb
Calculate Factorial
11:15
5
Calculate Factorial
Factorial Value: 120

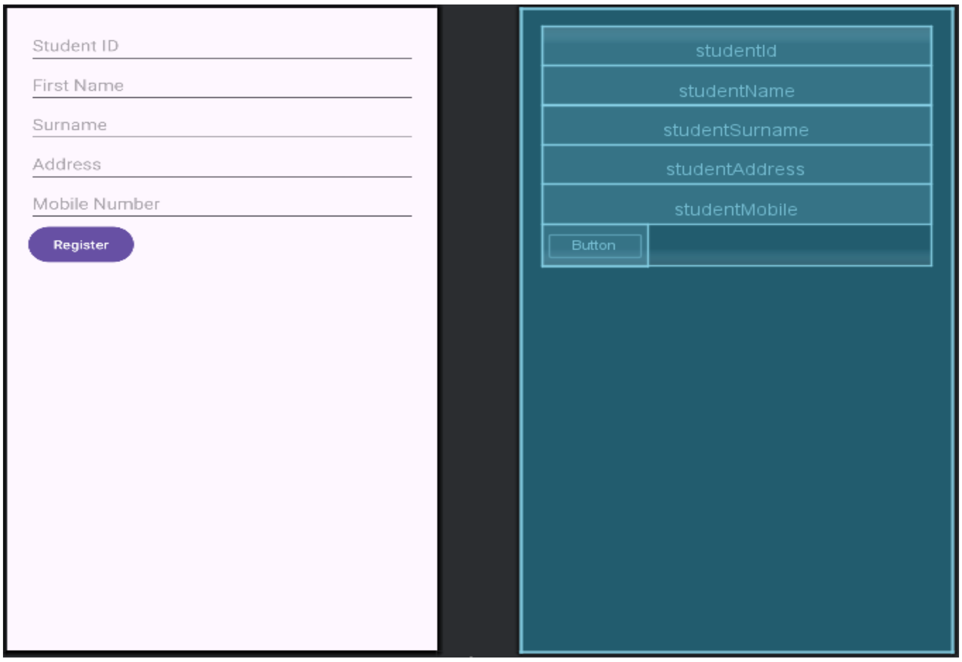
12. Develop a program to implement UI controls and create a Students Registration Form for College. (Include Student ID, Name, Surname, Address, Mobile No., etc) and display the details in another activity.

- **Program :-**

Layout/activity_main.xml

```
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical">

        <EditText
            android:id="@+id/studentId"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Student ID"
            android:inputType="number" />

        <EditText
            android:id="@+id/studentName"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="First Name"
            android:inputType="text" />

        <EditText
            android:id="@+id/studentSurname"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Surname"
            android:inputType="text" />

        <EditText
            android:id="@+id/studentAddress"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Address"
            android:inputType="text" />

        <EditText
```

```
            android:id="@+id/studentMobile"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Mobile Number"
            android:inputType="phone" />

        <Button
            android:id="@+id/registerButton"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Register" />
    </LinearLayout>
</ScrollView>
```



Layout/activity_display.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:id="@+id/displayText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:text="Student Details will appear here" />
</LinearLayout>
```

Java/com.example.appname/MainActivity.java

```
package com.example.studentregistration;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText studentId, studentName, studentSurname, studentAddress, studentMobile;
    Button registerButton;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        studentId = findViewById(R.id.studentId);
        studentName = findViewById(R.id.studentName);
        studentSurname = findViewById(R.id.studentSurname);
        studentAddress = findViewById(R.id.studentAddress);
        studentMobile = findViewById(R.id.studentMobile);
        registerButton = findViewById(R.id.registerButton);

        registerButton.setOnClickListener(v -> {
            Intent intent = new Intent(MainActivity.this, DisplayActivity.class);
```

```
            intent.putExtra("id", studentId.getText().toString());
            intent.putExtra("name", studentName.getText().toString());
            intent.putExtra("surname", studentSurname.getText().toString());
            intent.putExtra("address", studentAddress.getText().toString());
            intent.putExtra("mobile", studentMobile.getText().toString());
            startActivity(intent);
        });
    }
}
```

Java/com.example.appname/DisplayActivity.java

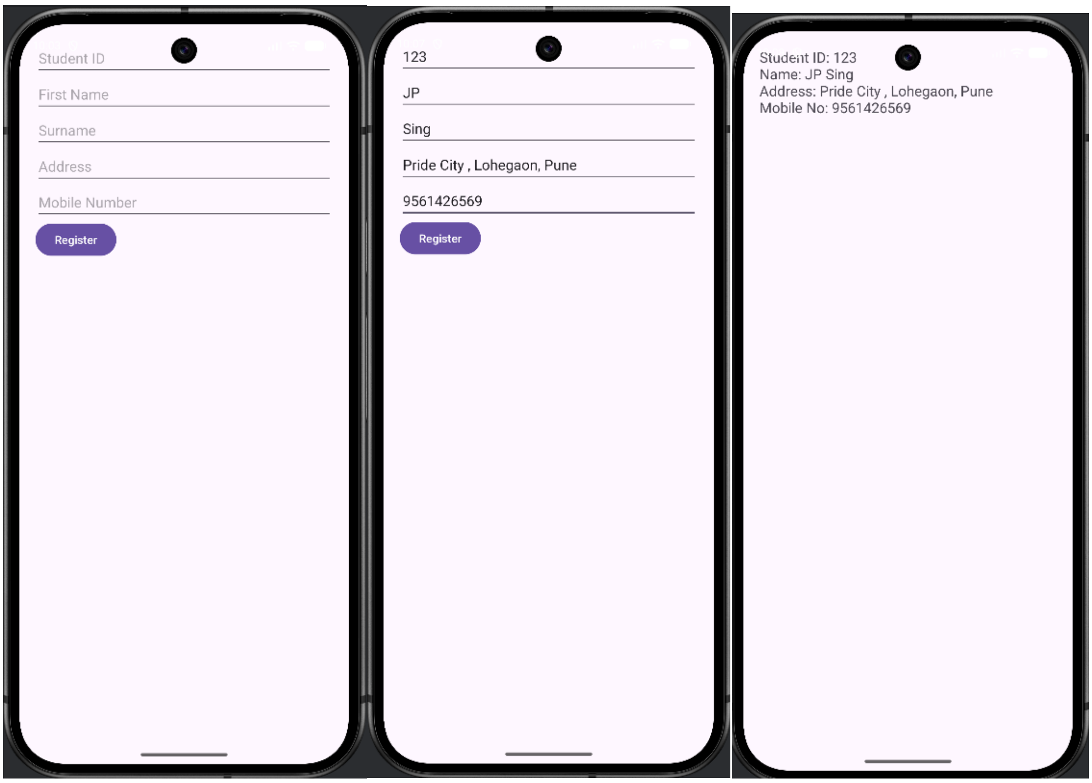
```
package com.example.studentregistration;

import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class DisplayActivity extends AppCompatActivity {

    TextView displayText;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_display);

        displayText = findViewById(R.id.displayText);

        String id = getIntent().getStringExtra("id");
        String name = getIntent().getStringExtra("name");
        String surname = getIntent().getStringExtra("surname");
        String address = getIntent().getStringExtra("address");
        String mobile = getIntent().getStringExtra("mobile");

        String details = "Student ID: " + id + "\n"
                + "Name: " + name + " " + surname + "\n"
                + "Address: " + address + "\n"
                + "Mobile No: " + mobile;

        displayText.setText(details);
    }
}
```

Manifest/AndroidMainfest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
```

```xml
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Studentregistration">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        <!-- Register Activity -->
        <activity android:name=".DisplayActivity"/>
    </application>

</manifest>
```

- **Output :-**

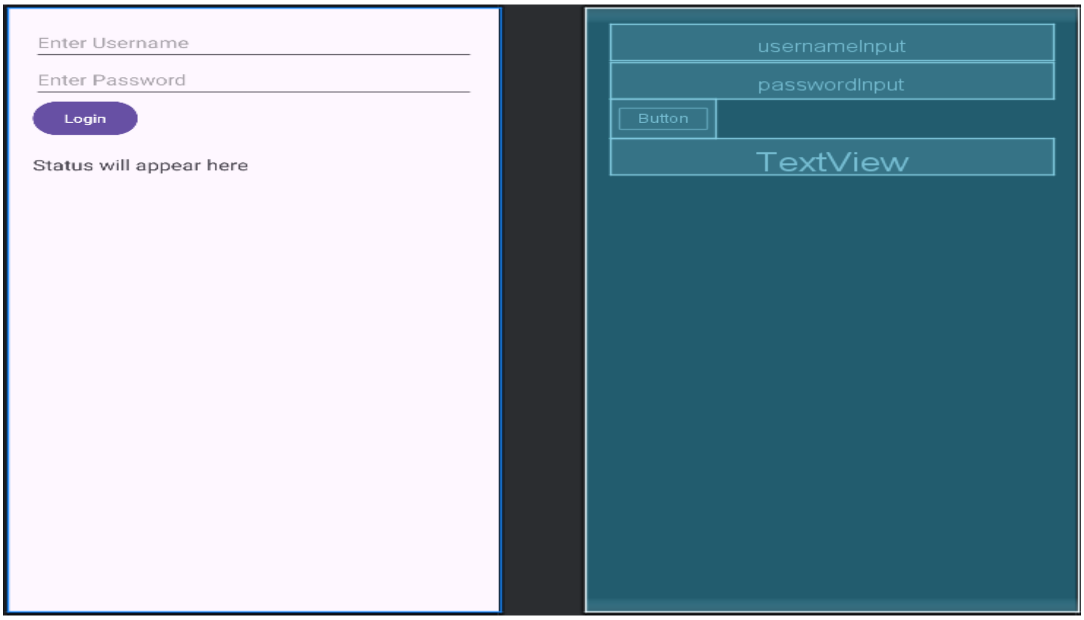


13. Create sample application with login module. (Check username and password) On successful login, Change TextView "Login Successful". And on login fail, alert user using Toast "Login fail".

- **Program :-**

Layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/usernameInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Username"
        android:inputType="textPersonName" />

    <EditText
        android:id="@+id/passwordInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Password"
        android:inputType="textPassword" />

    <Button
        android:id="@+id/loginButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Login" />

    <TextView
        android:id="@+id/loginStatus"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Status will appear here"
        android:textSize="18sp"
        android:paddingTop="20dp"/>
</LinearLayout>
```



Mainfests/AndroidMainfest.xml

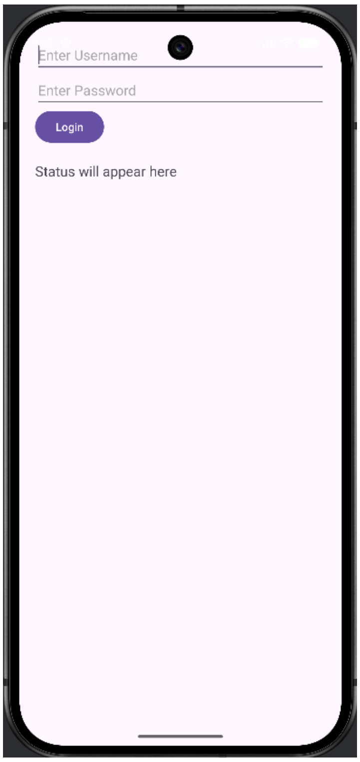
```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Loginapp">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
<!--        <activity android:name="."-->
    </application>

</manifest>
```

Java/com.example.appname/MainActivity.java

```
package com.example.loginapp;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

import java.util.HashMap;

public class MainActivity extends AppCompatActivity {

    EditText usernameInput, passwordInput;
    Button loginButton;
    TextView loginStatus;

    // Store multiple users in a HashMap
    private final HashMap<String, String> users = new HashMap<>();

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        usernameInput = findViewById(R.id.usernameInput);
        passwordInput = findViewById(R.id.passwordInput);
        loginButton = findViewById(R.id.loginButton);
        loginStatus = findViewById(R.id.loginStatus);


        // Add multiple user accounts
        users.put("admin", "1234");
        users.put("krushna", "abcd");
        users.put("student", "pass123");
        users.put("teacher", "teach2026");

        loginButton.setOnClickListener(v -> {
            String username = usernameInput.getText().toString();
            String password = passwordInput.getText().toString();


            if (users.containsKey(username) && users.get(username).equals(password)) {
                loginStatus.setText("Login Successful");
```

```
            } else {
                loginStatus.setText("Login Failed");
                Toast.makeText(MainActivity.this, "Login fail",
Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

- **Output :-**

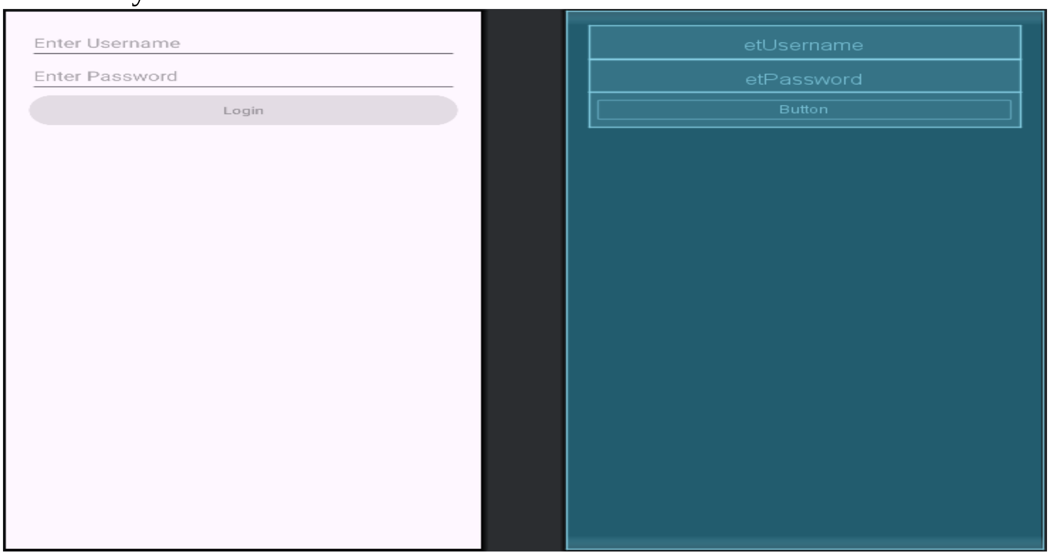


14. Create login application where you will have to validate username and password till the username and password is not validated, login button should remain disabled.

- **Program :-**

Layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <EditText
        android:id="@+id/etUsername"
        android:hint="Enter Username"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:inputType="textPersonName"/>
    <EditText
        android:id="@+id/etPassword"
        android:hint="Enter Password"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:inputType="textPassword"/>
    <Button
        android:id="@+id/btnLogin"
        android:text="Login"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:enabled="false"/> <!-- Disabled initially -->
</LinearLayout>
```



Java/com.example.app_name/MainActivity.java

```
package com.example.loginvalidation;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.text.Editable;
import android.text.TextWatcher;
import android.widget.Button;
import android.widget.EditText;

public class MainActivity extends AppCompatActivity {

    EditText etUsername, etPassword;
    Button btnLogin;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etUsername = findViewById(R.id.etUsername);
        etPassword = findViewById(R.id.etPassword);
        btnLogin = findViewById(R.id.btnLogin);

        TextWatcher watcher = new TextWatcher() {
            @Override
           public void beforeTextChanged(CharSequence s, int start, int count, int after) {}

            @Override
            public void onTextChanged(CharSequence s, int start, int before, int count) {
                validateInputs();
            }

            @Override
            public void afterTextChanged(Editable s) {}
        };

        etUsername.addTextChangedListener(watcher);
        etPassword.addTextChangedListener(watcher);
    }

    private void validateInputs() {
        String username = etUsername.getText().toString().trim();
        String password = etPassword.getText().toString().trim();

      // Example validation: username must be at least 3 chars, password at least 6 chars
```

```
        boolean isValid = username.length() >= 3 && password.length() >= 6;

        btnLogin.setEnabled(isValid);
    }
}


//Only when username ≥ 3 chars and password ≥ 6 chars, the button becomes enabled.

//asdf
//asdf123
```

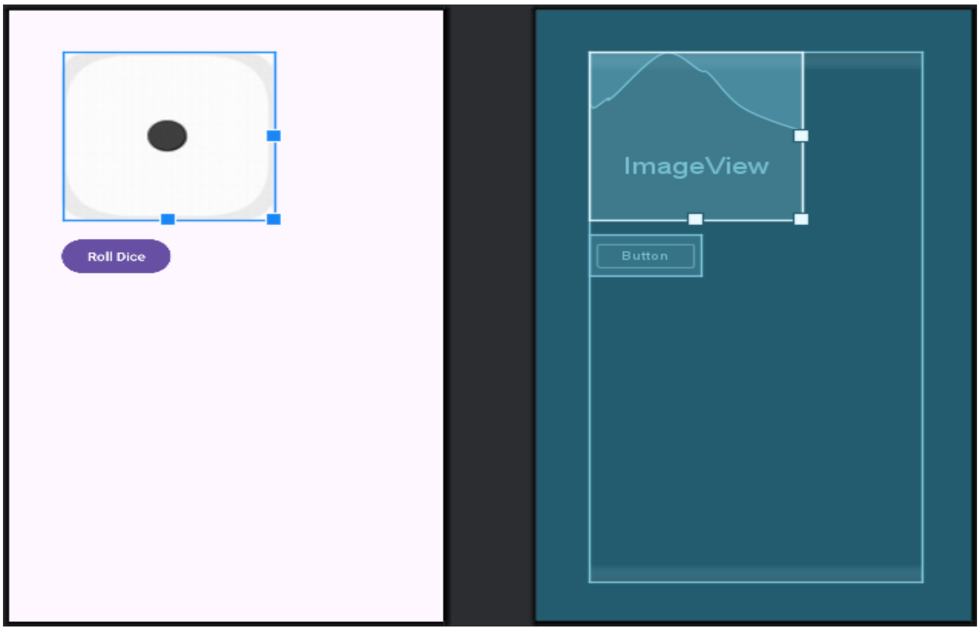
➢ Output :-

15. Develop any runnable Android Application.

- **Program :-**

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity"
    android:layout_margin="50dp"
    >

    <ImageView
        android:id="@+id/diceImage"
        android:layout_width="200dp"
        android:layout_height="200dp"
        android:src="@drawable/dice1" />

    <Button
        android:id="@+id/rollButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Roll Dice" />

</LinearLayout>
```

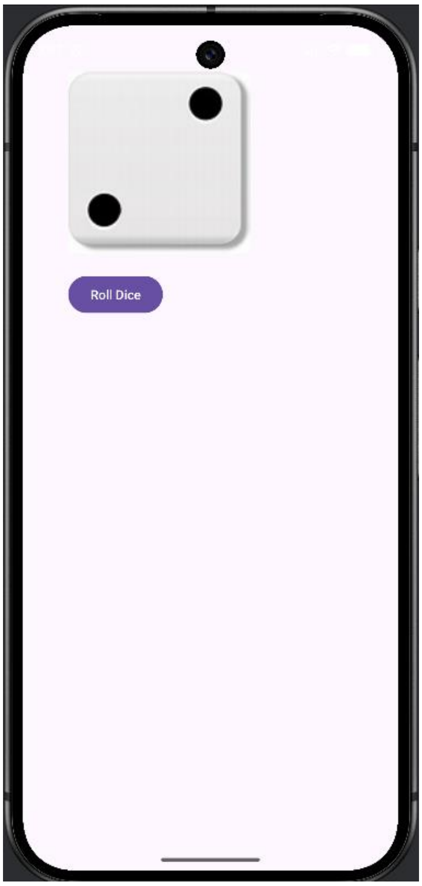


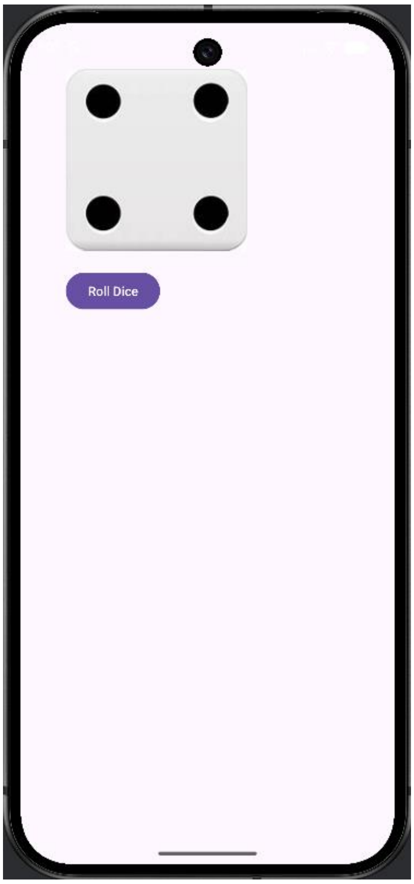
MainActivity.java

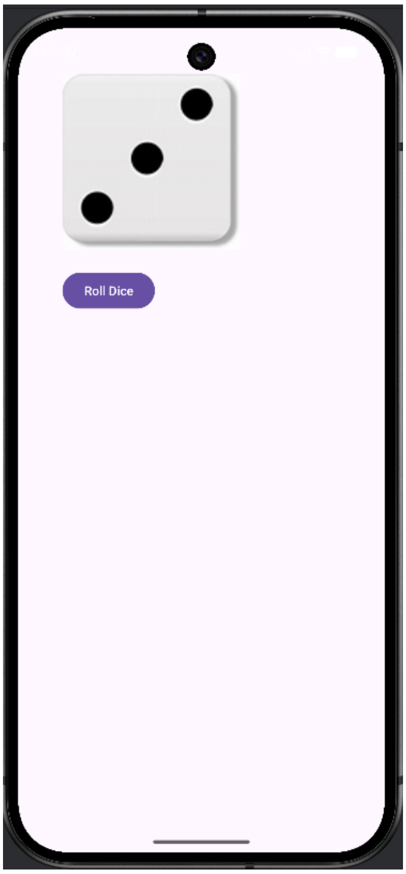
```java
package com.example.rolldice;

import android.os.Bundle;
import android.widget.Button;
import android.widget.ImageView;

import androidx.appcompat.app.AppCompatActivity;

import java.util.Random;

public class MainActivity extends AppCompatActivity {

    ImageView diceImage;
    Button rollButton;
    Random random;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        diceImage = findViewById(R.id.diceImage);
        rollButton = findViewById(R.id.rollButton);
        random = new Random();

        rollButton.setOnClickListener(v -> {
            int number = random.nextInt(6) + 1;

            switch (number) {
                case 1:
                    diceImage.setImageResource(R.drawable.dice1);
                    break;
                case 2:
                    diceImage.setImageResource(R.drawable.dice2);
                    break;
                case 3:
                    diceImage.setImageResource(R.drawable.dice3);
                    break;
                case 4:
                    diceImage.setImageResource(R.drawable.dice4);
                    break;
                case 5:
                    diceImage.setImageResource(R.drawable.dice5);
                    break;
                case 6:
```

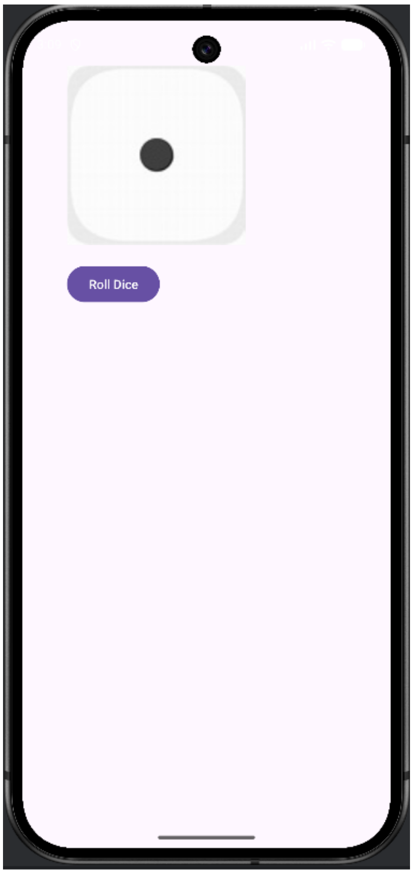
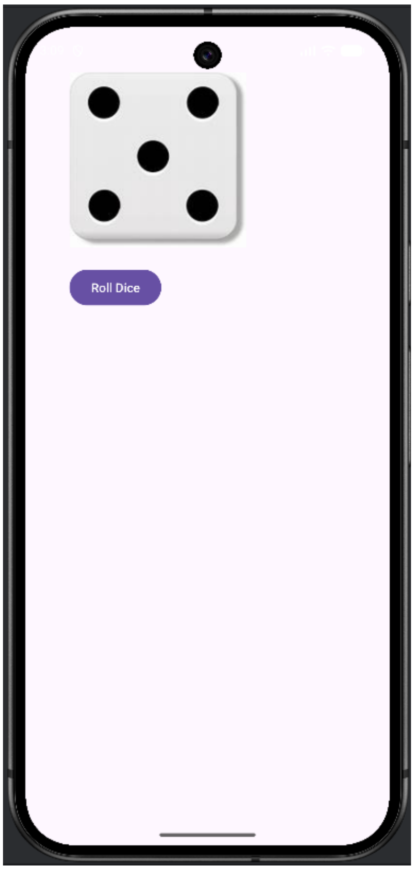
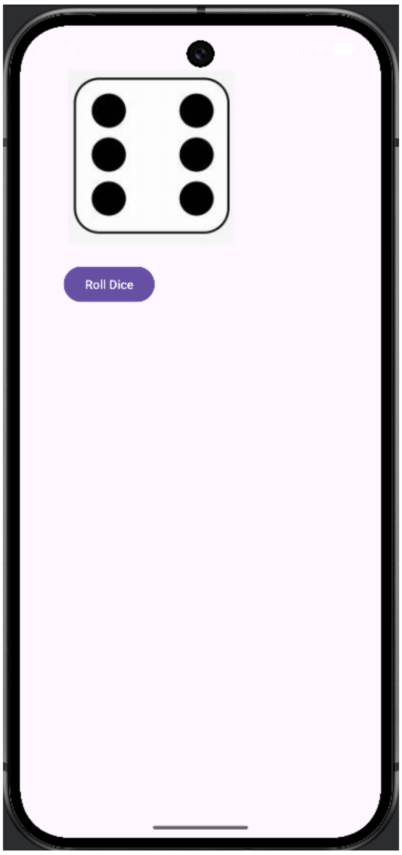
```
                diceImage.setImageResource(R.drawable.dice6);
                break;
            }
        });
    }
}
```

- **Output :-**








Roll Dice

Roll Dice

Roll Dice